

Ex# 12.4

Example #01

Find $u \times v$ and $v \times u$ if $u = 2i + j + k$ and $v = -4i + 3j + k$

Sol:-

$$u \cdot v = \begin{vmatrix} i & j & k \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} i - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} j + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} k$$

$$\begin{aligned} & [3(1) - 1(3)]i - [-4(1) - 2(1)]j + [2(3) + 4(1)]k \\ & |3 - 1| i - |-4 - 2| j + |6 + 4| k \\ & -2i + 6j + 10k \end{aligned}$$

$$\begin{aligned} v \times u &= -(u \times v) \\ &= -(-2i + 6j + 10k) \\ &= 2i - 6j - 10k \end{aligned}$$

Example #08

Find a vector perpendicular to the plane of
 $P(1, -1, 0)$ $Q(2, 1, -2)$ & $R(-1, 1, 2)$

Sol:

$$\vec{PQ} = (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ (2 - 1, 1 - (-1), -2 - 0) \\ (1, 2, -2)$$

Convert this into i, j, k form

$$\vec{PQ} = 1i + 2j - 2k$$

$$\vec{PR} = (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ = (-1 - 1, 1 - (-1), 2 - 0) \\ (-2, 2, 2)$$

$$\vec{PR} = -2i + 2j + 2k$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} i & j & k \\ 1 & 2 & -2 \\ -2 & 2 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -1 \\ -2 & 2 \end{vmatrix} i - \begin{vmatrix} 1 & -2 \\ -2 & 2 \end{vmatrix} j + \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix} k$$

$$\begin{vmatrix} 2(-1) - 2(2) \end{vmatrix} i - \begin{vmatrix} -2(-1) - 2(1) \end{vmatrix} j + \begin{vmatrix} -2(2) - 2(2) \end{vmatrix} k \\ \begin{vmatrix} -2 & -4 \end{vmatrix} i - \begin{vmatrix} 2 & -2 \end{vmatrix} j + \begin{vmatrix} -4 & -2 \end{vmatrix} k \\ -6i - 0 + -6k \\ -6i - 6k$$

Example #03

Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$ & $R(-1, 1, 2)$

Sol:-

The area of Parallelogram

$$|\vec{PQ} \times \vec{PR}| =$$

$$\begin{aligned}\vec{PQ} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (2 - 1, 1 - (-1), -1 - 0) \\ &= (1, 2, -1)\end{aligned}$$

$$\vec{PQ} = 1\mathbf{i} + 2\mathbf{j} - 1\mathbf{k}$$

$$\begin{aligned}\vec{PR} &= (-1 - 1, 1 - (-1), 2 - 0) \\ &= (-2, 2, 2) \\ &= -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}\end{aligned}$$

$$\cancel{1\mathbf{i} + 2\mathbf{j} - 1\mathbf{k}} \quad \cancel{(-2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})}$$

$$\cancel{-2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}}$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 2 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & -1 \\ -2 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix} \mathbf{k}$$

$$|4 + 2|\mathbf{i} - |2 - 2|\mathbf{j} + |4 - 2|\mathbf{k}$$

$$6\mathbf{i} - 0 + 2\mathbf{k} = 6\mathbf{i} + 2\mathbf{k}$$

Example # 06

find the volume of the box (parallelepiped) determined by $u = i + 2j - k$, $v = -2i + 3k$ and $w = 7j - 4k$.

Sol:-

$$(u \times v) \cdot w = \begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & 3 \\ 7 & -4 \end{vmatrix} - 2 \begin{vmatrix} -2 & 3 \\ 0 & 4 \end{vmatrix} + (-1) \begin{vmatrix} -2 & 0 \\ 0 & 7 \end{vmatrix}$$

$$1(0 \times (-4) - 7(3))$$

$$- 2(-4(-2) - 0) + (-1)(-2(7) - 0)$$

$$-21 - 16 + 14$$

$$= -23$$

The volume is -23

Problems

01- $u = 2i - 2j - k$

$v = i - k$

Sol:-

$$u \times v = \begin{vmatrix} i & j & k \\ 2 & -2 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & -1 \\ 0 & -1 \end{vmatrix} i - \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} j + \begin{vmatrix} 2 & -2 \\ 1 & 0 \end{vmatrix} k$$

$$= |-2(-1) - 0(-1)| i - |2(-1) - 1(-1)| j + |2(0) - 1(-2)| k$$

$$= |2 - 0| i - |-2 + 1| j + |0 + 2| k$$

$$= 2i - (-1)j + 2k$$

Length

$$|u \times v| = \sqrt{(2)^2 + (-1)^2 + (2)^2}$$

$$= \sqrt{4 + 1 + 4}$$

$$= \sqrt{9}$$

$$= 3$$

$$\text{Direction } (u \times v) = \frac{u \times v}{|u \times v|} = \frac{2i - j + 2k}{3}$$

$$\frac{2}{3}i - \frac{1}{3}j + \frac{2}{3}k$$

Continue

Now find $v \times u$

$$\begin{aligned}(v \times u) &= -(u \times v) \\ &= -(2i + j + 2k) \\ &= -2i - j - 2k\end{aligned}$$

$$\begin{aligned}\text{Length} &= \sqrt{(-2)^2 + (-1)^2 + (-2)^2} \\ &= \sqrt{4 + 1 + 4} \\ &= \sqrt{9} \\ &= 3\end{aligned}$$

$$\text{Direction} = \frac{v \times u}{|v \times u|} = \frac{-2i - j - 2k}{3}$$

$$= -\frac{2}{3}i - \frac{1}{3}j - \frac{2}{3}k$$

12) $u = 2i + 3j$, $v = -i + j$

Sol:-

$$u \times v = \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ -1 & 1 & 0 \end{vmatrix}$$

$$\Rightarrow u \times v = \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ -1 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 0 \\ 1 & 0 \end{vmatrix} i - \begin{vmatrix} 2 & 0 \\ -1 & 0 \end{vmatrix} j + \begin{vmatrix} 2 & 3 \\ -1 & 1 \end{vmatrix} k$$

$$\begin{vmatrix} 3(0) - 1(0) \end{vmatrix} i - \begin{vmatrix} 2(0) - (-1(0)) \end{vmatrix} j + \begin{vmatrix} 2(1) + 1(3) \end{vmatrix} k$$

$$0 - 0 + \{2 + 3\}k$$

$$0 + 5k$$

$$\text{Length} = \sqrt{(5)^2} = 5$$

$$\text{Direction} = \frac{5k}{5} = k$$

$$(v \times u) = -(u \times v)$$

$$= -5$$

$$\text{Length} : \sqrt{(-5)^2} = 5$$

$$\text{Direction} : -k$$

03) $u = 2i - 2j + 4k$, $v = -i + j - 2k$

Sol:-

$$u \times v = \begin{vmatrix} i & j & k \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 4 \\ 1 & -2 \end{vmatrix} i - \begin{vmatrix} 2 & 4 \\ -1 & -2 \end{vmatrix} j + \begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix} k$$

$$\begin{aligned} & |(-2)(-2) - 4(1)| i - |2(-2) - 4(-1)| j + |2(1) - (-2(-1))| k \\ & |4 - 4| i - |-4 - (-4)| j + |2 - 2| k \\ & 0 \quad \quad \quad 0 \quad \quad \quad + \quad \quad 0 \end{aligned}$$

Length = 0

$v \times u = 0$

04) $u = i + j - k$ $v = 0$

Sol:-

$$u \times v = \begin{vmatrix} i & j & k \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} i - \begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} j + \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} k$$

$$\begin{matrix} |1(0) - (-1(0))| i & - & |1(0) + 1(0)| j & + & |1(0) - 1(0)| k \\ 0 & & 0 & & 0 \end{matrix}$$

Length : 0

Direction : NO

$$v \times u = 0$$

Triangles in Space

15. $P(1, -1, 2)$, $Q(2, 0, -1)$, $R(0, 2, 1)$

Sol:-

$$(\vec{P} \times \vec{Q}) \cdot \vec{R} = \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ 2 & 0 & -1 \\ 0 & 2 & 1 \end{vmatrix}$$

a) Find the area of triangle PQR

$$\frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$\begin{aligned} \vec{PQ} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (2 - 1, 0 - (-1), -1 - 2) \\ &= (1, 1, -3) \end{aligned}$$

$$\begin{aligned} \vec{PR} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (0 - 1, 2 - (-1), 1 - 2) \\ &= (-1, 3, -1) \end{aligned}$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} i & j & k \\ 1 & 1 & -3 \\ -1 & 3 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -3 \\ 3 & -1 \end{vmatrix} i - \begin{vmatrix} 1 & -3 \\ -1 & -1 \end{vmatrix} j + \begin{vmatrix} 1 & 1 \\ -1 & 3 \end{vmatrix} k \quad |1(3) + 1$$

$$\begin{aligned} &= |1(-1) - 3(-3)|i - |1(-3) + 1(-1)|j + |1(3) - 3(-1)|k \\ &= |-1 + 9|i - |-3 - 1|j + |3 + 3|k \\ &= 8i - 4j + 6k \end{aligned}$$

Area

$$\frac{1}{2} |\vec{PO} \times \vec{PR}|$$

$$\Rightarrow \frac{1}{2} \sqrt{(8)^2 + (4)^2 + (4)^2}$$

$$\Rightarrow \frac{1}{2} \sqrt{64 + 16 + 16}$$

$$2\sqrt{6}$$

B) find a unit vector perpendicular

$$\frac{\vec{PO} \times \vec{PR}}{|\vec{PO} \times \vec{PR}|} = \frac{8\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}}{2\sqrt{6}}$$

cos θ .

cos $\frac{2\pi}{3}$

16) P(1, 1, 1) Q(2, 1, 3) R(3, -1, 1)

Sol:-

$$\vec{PQ} = (2-1, 1-1, 3-1) \\ = (1, 0, 2)$$

$$\vec{PR} = (3-1, -1-1, 1-1) \\ = (2, -2, 0)$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 2 & -2 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 2 & -2 \end{vmatrix} \mathbf{k}$$

$$= |0(0) + 2(2)| \mathbf{i} - |1(0) - 2(2)| \mathbf{j} + |1(-2) - 2(0)| \mathbf{k} \\ = |0 + 4| \mathbf{i} - |0 - 4| \mathbf{j} + |-2 - 0| \mathbf{k} \\ = 4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$$

$$\text{Area} = \frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$= \frac{1}{2} \sqrt{(4)^2 + (4)^2 + (-2)^2} \Rightarrow \frac{1}{2} \sqrt{16 + 16 + 4}$$

$$\frac{1}{2} \sqrt{36} = \frac{6}{2} = 3$$

$$\text{b) } \frac{\vec{PQ} \times \vec{PR}}{|\vec{PQ} \times \vec{PR}|} = \frac{4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}}{6}$$

$$17) P(2, -2, 1) \quad Q(3, -1, 2) \quad R(3, -1, 1)$$

Sol:-

$$\vec{PQ} = (3-2, -1+2, 2-1) \\ = (1, 1, 1)$$

$$\vec{PR} = (3-2, -1+2, 1-1) \\ = (1, 1, 0)$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} \hat{i} - \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} \hat{j} + \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \hat{k}$$

$$|1(0) - 1(1)|\hat{i} - |1(0) - 1(1)|\hat{j} + |1(1) - 1(1)|\hat{k} \\ = -1\hat{i} + 1\hat{j} + 0$$

$$\text{Area} = \frac{1}{2} |\vec{PQ} \times \vec{PR}| \Rightarrow \frac{1}{2} |-\hat{i} + \hat{j}|$$

$$\frac{1}{2} \sqrt{(-1)^2 + 1^2} \Rightarrow \frac{\sqrt{2}}{2}$$

$$B) \quad u = \frac{\vec{PQ} \times \vec{PR}}{|\vec{PQ} \times \vec{PR}|} = \frac{-\hat{i} + \hat{j}}{\frac{\sqrt{2}}{2}} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}}$$

$$18) P(-2, 2, 0) \quad Q(0, 1, -1) \quad R(-1, 2, -2)$$

Sol:-

$$\begin{aligned}\vec{PQ} &= (0+2, 1-2, -1-0) \\ &= (2, -1, -1)\end{aligned}$$

$$\begin{aligned}\vec{PR} &= (-1+2, 2-2, -2-0) \\ &= (1, 0, -2)\end{aligned}$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -1 \\ 1 & 0 & -2 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -1 \\ 0 & -2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} \mathbf{k}$$

$$\begin{aligned}&= (-1(-2) + 1(0))\mathbf{i} - (2(-2) + 1(1))\mathbf{j} + (2(0) + 1(1))\mathbf{k} \\ &= (2 + 0)\mathbf{i} - (-4 + 1)\mathbf{j} + (0 + 1)\mathbf{k} \\ &= 2\mathbf{i} + 3\mathbf{j} + 1\mathbf{k}\end{aligned}$$

$$\text{Area} = \frac{1}{2} |\vec{PQ} \times \vec{PR}| \Rightarrow \frac{1}{2} \sqrt{2^2 + 3^2 + 1^2}$$

$$\frac{1}{2} \sqrt{4+9+1} \Rightarrow \frac{1}{2} \sqrt{14} \Rightarrow \frac{\sqrt{14}}{2}$$

$$B) \mathbf{u} = \frac{\vec{PQ} \times \vec{PR}}{|\vec{PQ} \times \vec{PR}|} = \frac{2\mathbf{i} + 3\mathbf{j} + \mathbf{k}}{\sqrt{14}}$$

Area of Parallelogram

35) A(1,0), B(0,1), C(-1,0), D(0,-1)

find vectors according to Adjacency

AB and AD

These are adjacency vertices

$$\vec{AB} = (0-1, 1-0) \\ (-1, 1)$$

$$\vec{AD} = (0-1, -1-0) \\ (-1, -1)$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -1 & -1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} i - \begin{vmatrix} -1 & 0 \\ -1 & 0 \end{vmatrix} j + \begin{vmatrix} -1 & 1 \\ -1 & -1 \end{vmatrix} k$$

$$|1(0) + 1(0)|i - |-1(0) + 1(0)|j + |-1(-1) + 1(1)|k$$

$$0 - 0 + |1 + 1|k$$

$$2k$$

$$\text{Area} = \frac{1}{2} |\vec{AB} \times \vec{AD}| \Rightarrow \frac{1}{2} |2k|$$

$$\text{Area} = \frac{2}{2} \Rightarrow \sqrt{2^2} = 2$$

36) $A(0,0)$ $B(7,3)$ $C(9,8)$ $D(2,5)$

Sol:

$$\vec{AB} = (7-0, 3-0) = (7, 3)$$

$$\vec{AD} = (2-0, 5-0) = (2, 5)$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} i & j & k \\ 7 & 3 & 0 \\ 2 & 5 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 0 \\ 5 & 0 \end{vmatrix} i - \begin{vmatrix} 7 & 0 \\ 2 & 0 \end{vmatrix} j + \begin{vmatrix} 7 & 3 \\ 2 & 5 \end{vmatrix} k$$

$$= |3(0) - 5(0)|i - |7(0) - 2(0)|j + |7(5) - 2(3)|k$$

$$= 0i - 0j + 35 - 6k$$

$$= 29k$$

$$\text{Area} = \sqrt{(29)^2} = \sqrt{841} = 29$$

$$37) A(-1, 2) \quad B(2, 0) \quad C(7, 1) \quad D(4, 3)$$

Sol:-

$$\vec{AB} = (2+1, 0-2) \\ (3, -2)$$

$$\vec{AD} = (4+1, 3-2) \\ (5, 1)$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} i & j & k \\ 3 & -2 & 0 \\ 5 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} -2 & 0 \\ 1 & 0 \end{vmatrix} i - \begin{vmatrix} 3 & 0 \\ 5 & 0 \end{vmatrix} j + \begin{vmatrix} 3 & -2 \\ 5 & 1 \end{vmatrix} k$$

$$\begin{matrix} -2(0) - 1(0) & - & 3(0) - 5(0) & + & 3(1) + 2(5) \\ 0 & & 0 & & 3 + 10 \\ & & & & 13 \end{matrix} k$$

$$\text{Area} = \sqrt{(13)^2} = 13$$